Inverting Amplifier

-AIM

To design and analyze an inverting amplifier using an operational amplifier and verify the gain.

Working Principle: Inverting Amplifier

The Inverting Amplifier uses negative feedback to amplify an input signal while reversing its phase. The operation is defined by the "Virtual Ground" concept.

1. The Virtual Ground Concept

- The non-inverting terminal (V_+) is connected to **Ground (0V)**.
- Due to the high open-loop gain of the op-amp, it maintains the inverting terminal (V_-) at the same potential.
- This makes the inverting terminal a **Virtual Ground (0V)**.

2. Current Analysis (KCL)

Since the input impedance of the op-amp is nearly infinite, no current flows into the inverting input. Therefore, the current flowing through R_1 must equal the current flowing through R_f.

1. **Input Current (I_{in}):** $I_{in} = \frac{V_{in} - V_-}{R_1} = \frac{V_{in} - 0}{R_1} = \frac{V_{in}}{R_1}$
2. **Feedback Current (I_f):** $I_f = \frac{V_- - V_{out}}{R_f} = \frac{0 - V_{out}}{R_f} = -\frac{V_{out}}{R_f}$

3. Deriving the Gain

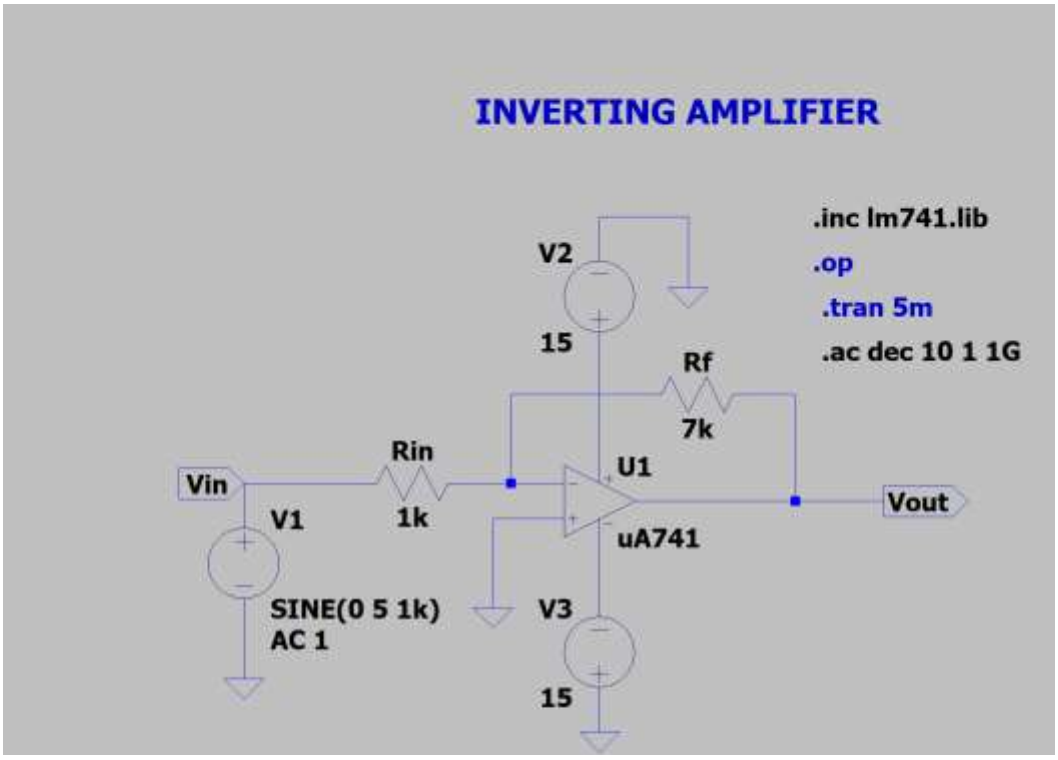
By equating the currents ($I_{in} = I_f$): $\frac{V_{in}}{R_1} = -\frac{V_{out}}{R_f}$

Rearranging for the Output Voltage: $V_{out} = -V_{in}\left(\frac{R_f}{R_1}\right)$

The Voltage Gain (A_V) is: $A_V = \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_1}$

The negative sign indicates a **180° phase inversion**, meaning the output signal is the mirror image of the input signal across the 0V axis.

Circuit diagram



Design part

given :

AV = -7v/v

Inverting Amplifier Design

Target Gain: $A_V = -7$ V/V

1. Design Calculations

The voltage gain for an inverting amplifier is given by the formula: $A_V = \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_1}$

Given: $A_V = -7$

Derivation: $-7 = -\frac{R_f}{R_1} R_f = 7 \cdot R_1$

Assumed Values:

- Let $R_1 = 1\text{k}\Omega$
- Therefore, $R_f = 7\text{k}\Omega$

DC Analysis

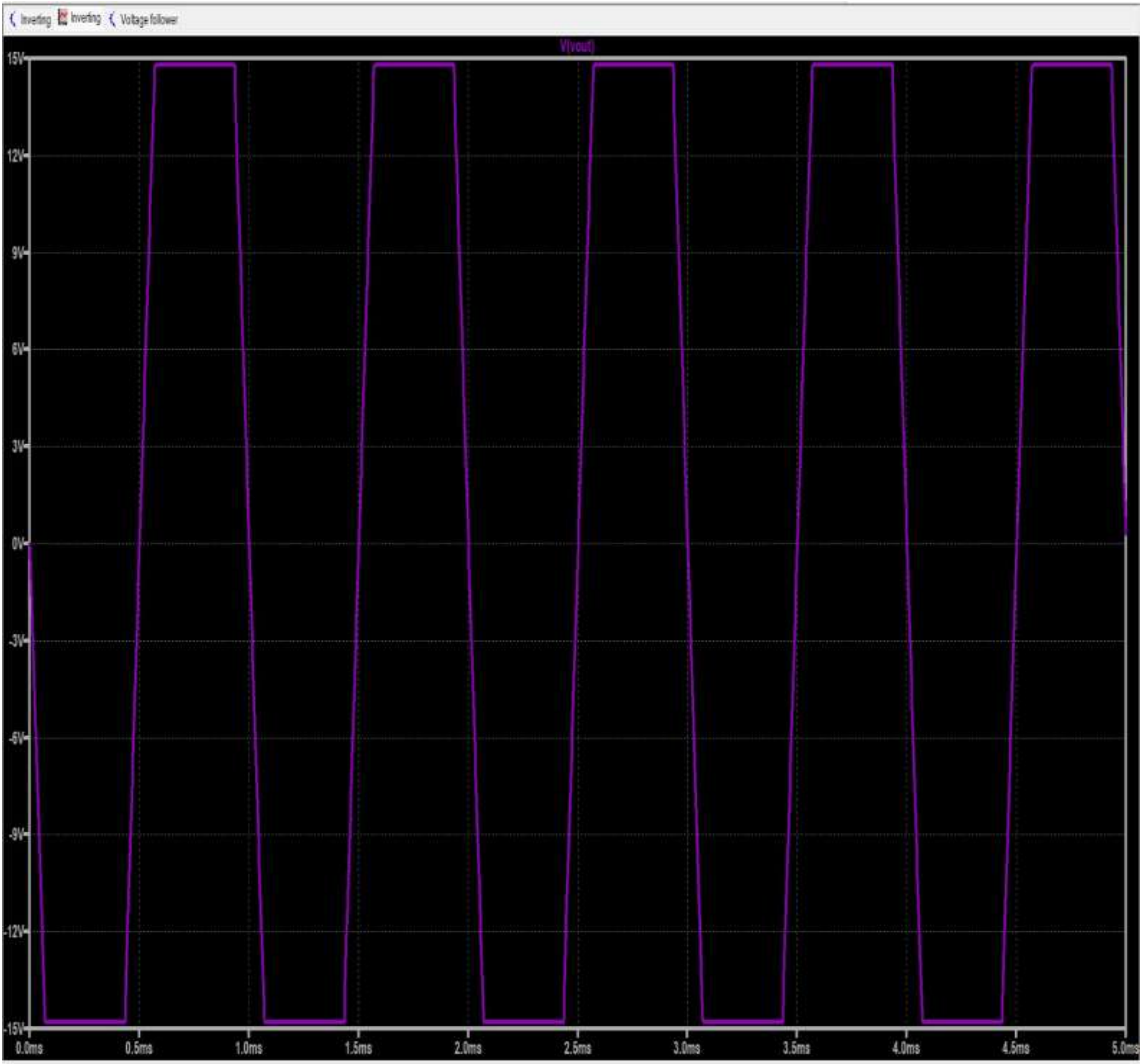
Transient Analysis

C:\Users\Vanshikha Muthanna C\OneDrive\Desktop\180nm\Inverting.net

```
       --- Operating Point ---

V(n001):        1.92485e-05    voltage
V(n002):        15             voltage
V(n003):        -15            voltage
V(vin):         0              voltage
V(vout):        0.000712321    voltage
I(Rf):          9.90104e-08    device_current
I(Rin):         1.92485e-08    device_current
I(V1):          1.92485e-08    device_current
I(V2):          -0.00166694    device_current
I(V3):          -0.00166714    device_current
Ix(u1:1):       7.97232e-08    subckt_current
Ix(u1:2):       7.97618e-08    subckt_current
Ix(u1:3):       0.00166694     subckt_current
Ix(u1:4):       -0.00166714    subckt_current
Ix(u1:5):       -9.90084e-08   subckt_current
```

Transient analysis:

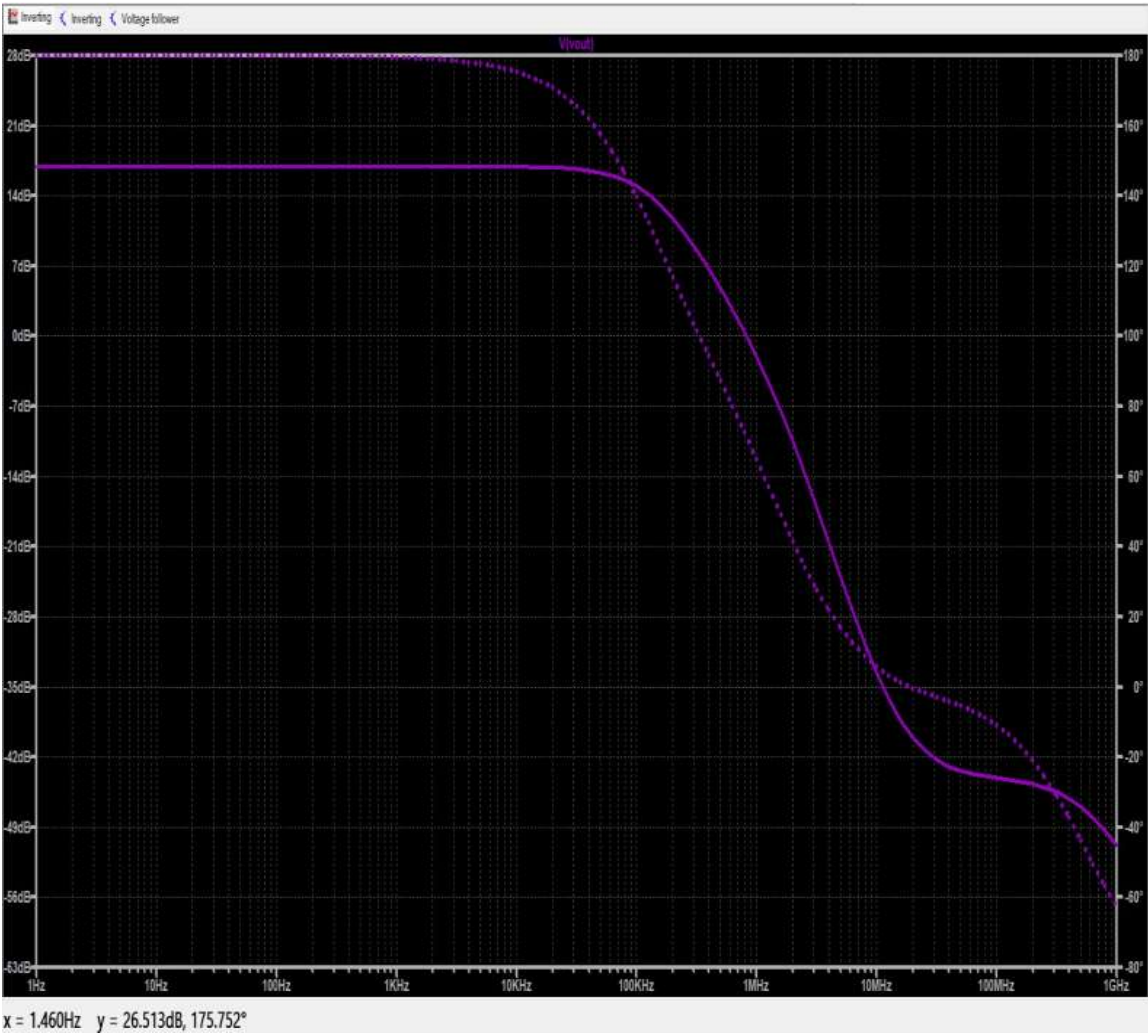

Inverting
Inverting
Voltage follower
V(vout)
15V
12V
9V
6V
3V
0V
-3V
-6V
-9V
-12V
-15V
0.0ms
0.5ms
1.0ms
1.5ms
2.0ms
2.5ms
3.0ms
3.5ms
4.0ms
4.5ms
5.0ms

output calculation

Vin(max) = 6 V Vout = 7 × 5 = 42 V Since supply voltage is ±15V:

Output cannot reach 55 V Output will saturate at ≈ ±13.5 V Clipping occurs because the required output voltage exceeds the supply voltage This causes distortion in the output waveform

AC Analysis



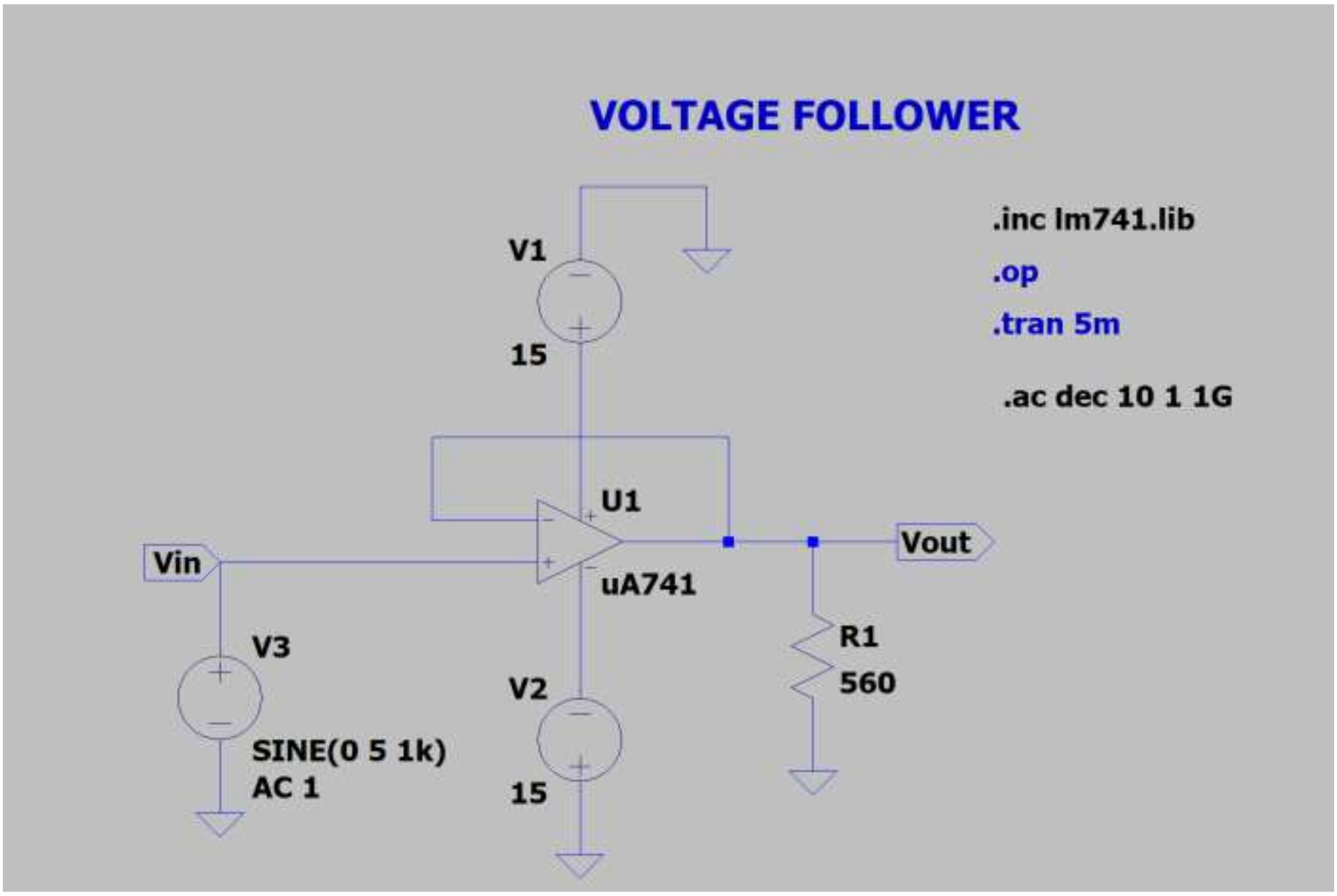
At low frequency: Gain ≈ 16.9 dB (since Av = 7) Gain remains constant up to bandwidth After cut-off frequency: Gain decreases

Voltage follower

Aim

To design and analyze a voltage follower circuit using an operational amplifier and verify that the output voltage follows the input voltage (unity gain).

Circuit diagram



A voltage follower is a special case of a non-inverting amplifier where the output is directly fed back to the inverting terminal.

General gain formula of non-inverting amplifier: Av = 1 + (Rf / R1)

For voltage follower: Rf = 0 and R1 = ∞

So, Av = 1

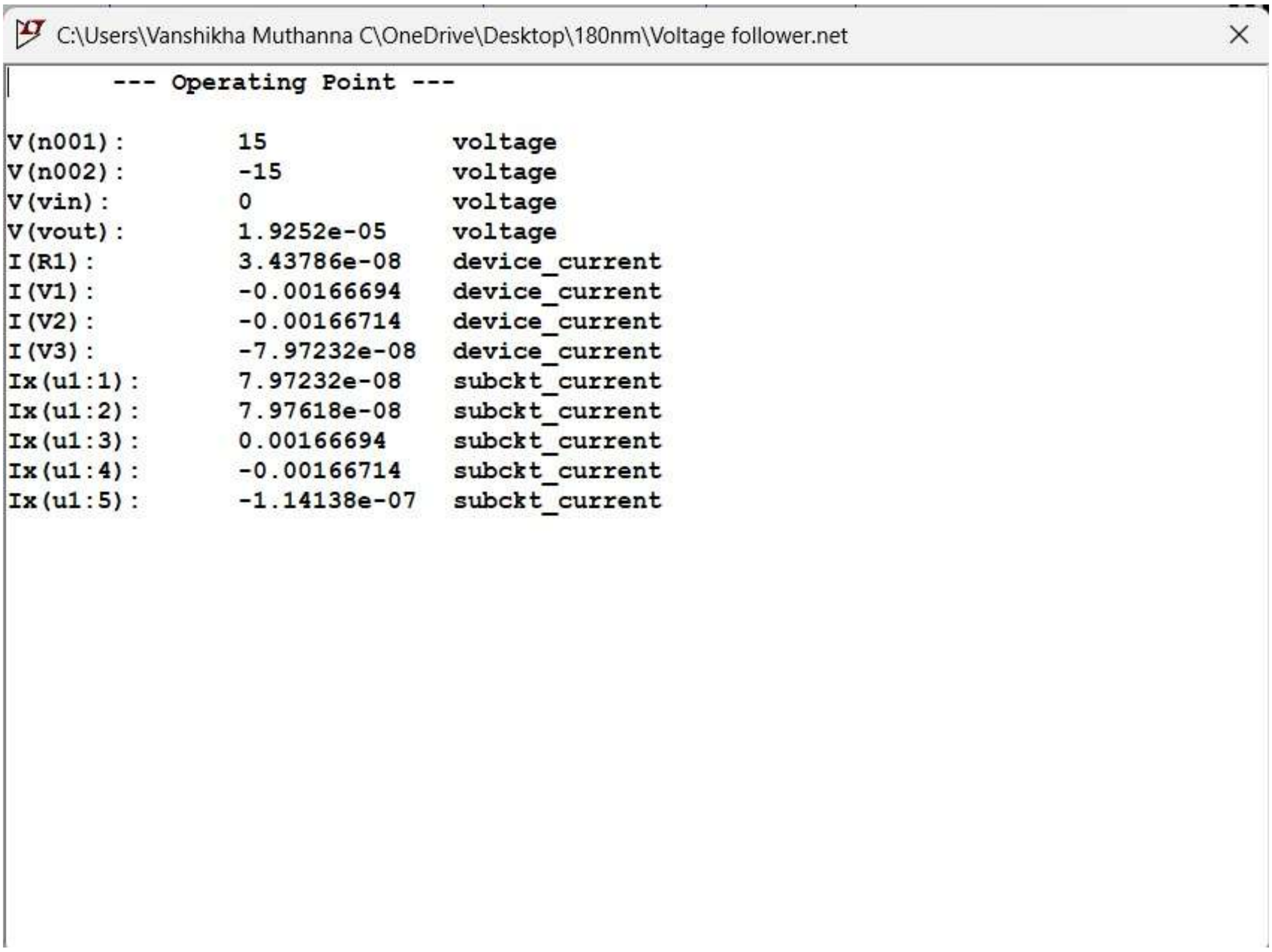
Therefore, Vout = Vin

Input is applied to non-inverting terminal (+) Output is directly connected to inverting terminal (−) No resistors are used -Device current → current through simple elements (R, V, etc.) -Subckt current → current entering/leaving a subcircuit block .

DC Analysis

Given: Vin(p-p) = 10 V Vcc = 15V RL = 5.6kΩ

Gain: Av = 1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--- Operating Point ---

V(n001):	15	voltage
V(n002):	-15	voltage
V(vin):	0	voltage
V(vout):	1.9252e-05	voltage
I(R1):	3.43786e-08	device_current
I(V1):	-0.00166694	device_current
I(V2):	-0.00166714	device_current
I(V3):	-7.97232e-08	device_current
Ix(u1:1):	7.97232e-08	subckt_current
Ix(u1:2):	7.97618e-08	subckt_current
Ix(u1:3):	0.00166694	subckt_current
Ix(u1:4):	-0.00166714	subckt_current
Ix(u1:5):	-1.14138e-07	subckt_current

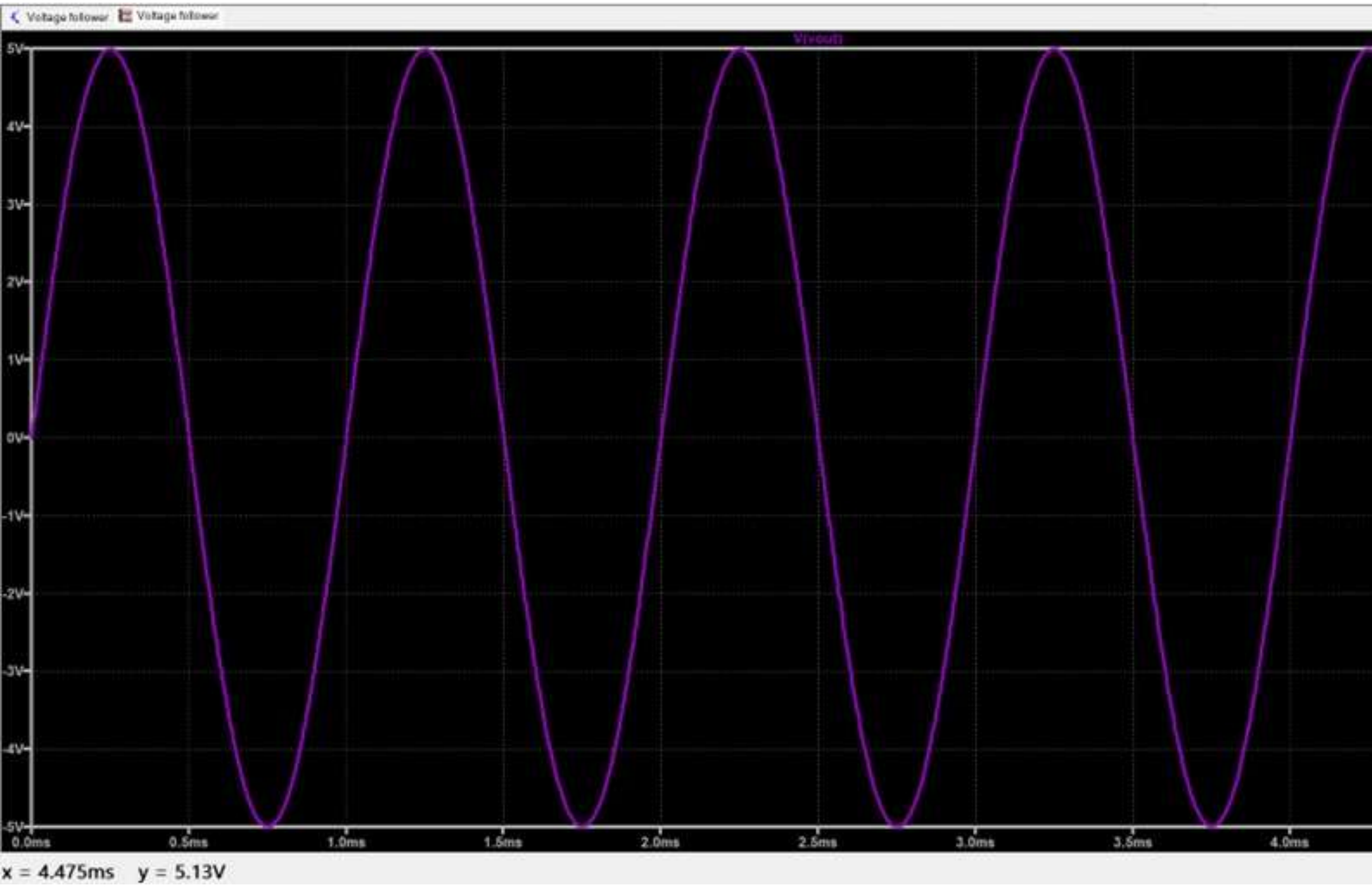
Transient analysis:

Output Signal Output waveform is also sinusoidal Amplitude: 5 V Peak-to-peak voltage: 10 V Frequency: Same as input (1 kHz) Vout = Vin

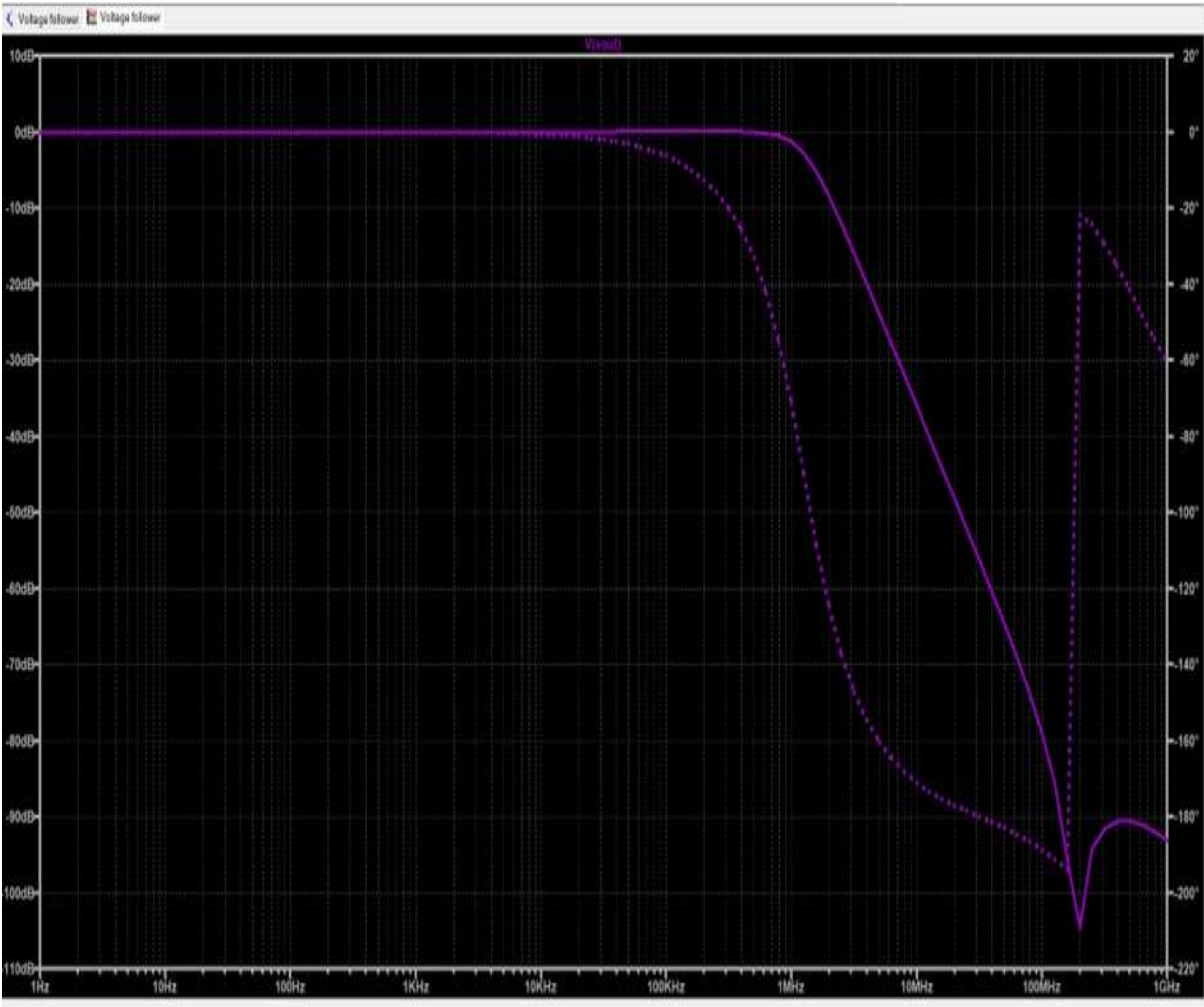
Type: Sinusoidal wave Amplitude: 5 V Peak-to-peak voltage: 10 V Frequency: 1 kHz Vin = 5 sin(2πft)

No Phase Shift Input and output peaks occur at the same time

No Clipping Output remains within ±5 V Supply voltage (±13 V) is sufficient



AC Analysis:



Reason for Dip in Frequency Response

Explanation

The dip observed in the frequency response graph is attributed to the **non-ideal characteristics** of the **uA741 op-amp model**.

- **Internal Poles and Zeros:** In a real-world or modeled uA741, internal transistors and compensation networks create multiple poles and zeros. At specific frequencies, these elements interact with one another.
- **Gain Reduction:** This interaction results in a localized loss of magnitude, causing a **temporary drop in gain (dip)** before the final high-frequency roll-off occurs.

Voltage Follower (Unity Gain Buffer)

Analysis

The voltage follower is characterized by **unity gain** ($A_V = 1$), meaning the output voltage directly tracks the input voltage. Its primary function is to serve as a **buffer** between stages of a circuit.

Key Characteristics

- **Impedance Transformation:** It features **high input impedance**, which prevents loading of the preceding stage, and **low output impedance**, allowing it to drive subsequent loads efficiently.
- **Signal Integrity:** By acting as an isolation layer, it ensures accurate signal transfer and prevents voltage drop across the source's internal resistance.
- **Phase Retention:** The output remains in phase with the input, providing a 1:1 voltage replica.